WalkSafe

Navigation Assistant for Visually Impaired Persons in Dense Urban Environments



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02 August 2026

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Project Overview: The Gap

“Visually impaired people in dense urban environments face moment-to-moment dangers that no current assistive tool is designed to handle completely.”

01

The Environment Problem

Streets in Dhaka are unplanned and unpredictable. Tactile paving, wide pavements, and clear pedestrian zones do not exist in most areas.

02

The Tool Problem

Existing tools like White Cane could reach only to 1-2 steps ahead. It fails if there’s a moving object. Also GPS system could help with navigation but can’t detect obsacles in real time.

03

The Verification Problem

A blind user cannot glance up and see the guidance is wrong. Every instruction is taken on faith, and a false "path clear" is undetectable until impact.

Project Overview: WalkSafe



Standard Camera

Live video feed



Smart Pipeline

Perception + Decision



Spoken Guidance

Via earphone

What the user hears:

- *Move forward — path clear”*
- *“Stop — person approaching from your right”*
- *“Step left — obstacle ahead at 1 metre”*
- *“Caution — low object above, slow down”*

Positioning: a supplementary aid. Every requirement assumes the user also has a cane or guide dog.

Project Overview: Scope

In Scope

- Smartphone camera, remote GPU service
- Depth, agent tracking, motion
- Distance, time-to-collision, risk per agent
- Walkable openings, measured in metres
- Nine commands — speech, earcon, haptic
- On-phone check — offline, overrides server
- Session logging of derived metrics + analytics view

Out of Scope

- Route planning, maps, destinations
- Reading text, signs, faces
- Traffic signals, road crossing
- Kerbs, stairs, drop-offs — declared limit
- Indoor mapping
- Replacing cane, guide dog, or O&M training

Functional Requirements

01 Capture and Perception

02 Interpretation

03 Navigation Decision

04 Guidance Delivery

05 Safety and Failure Handling

06 Setup, Session, Accessibility

07 Logging and Review

Total 26 Functional Requirements covered under 7 Areas

Functional Requirements

- FR-2 — Estimate distance to obstacles and to the ground ahead
- FR-5 — Distance, approach speed and time-to-collision per agent, from measured frame times
- FR-9 — Exactly one command per cycle from the nine, by fixed-order rules, cautious default
- FR-12 — Speak each command, play its earcon, vibrate for urgent ones
- FR-19 — Announce that guidance has stopped on any fault. Never fail silently
- FR-23 — Spoken setup: FOV, mounting, depth check, and a left/right test that blocks use if it fails
- FR-24 — Record every decision with its values. Store no images due to privacy concern

Non-Functional Requirements

01 Safety

- Every failure leaves the system more cautious — never less (NFR-5)
- Left and right are never reversed (NFR-6)
- No fault is signalled by silence alone (NFR-7)

04 Performance

A hard deadline for phone warnings, a looser one for server guidance; late guidance is discarded, not spoken (NFR-1–4)

05 Reliability

A long session runs without a crash or an unannounced loss of capability; no single dependency failure ends a session (NFR-9, 10)

06 Accessibility

Everything reachable without sight; earphones optional, because the user must hear the street (NFR-11–13)

07 Resources

Bounded battery and data; every threshold in one config file, models swappable without touching the rules (NFR-14–16)

Total 16 Functional Requirements covered under 5 Areas

Commands and Levels

Command Vocabulary

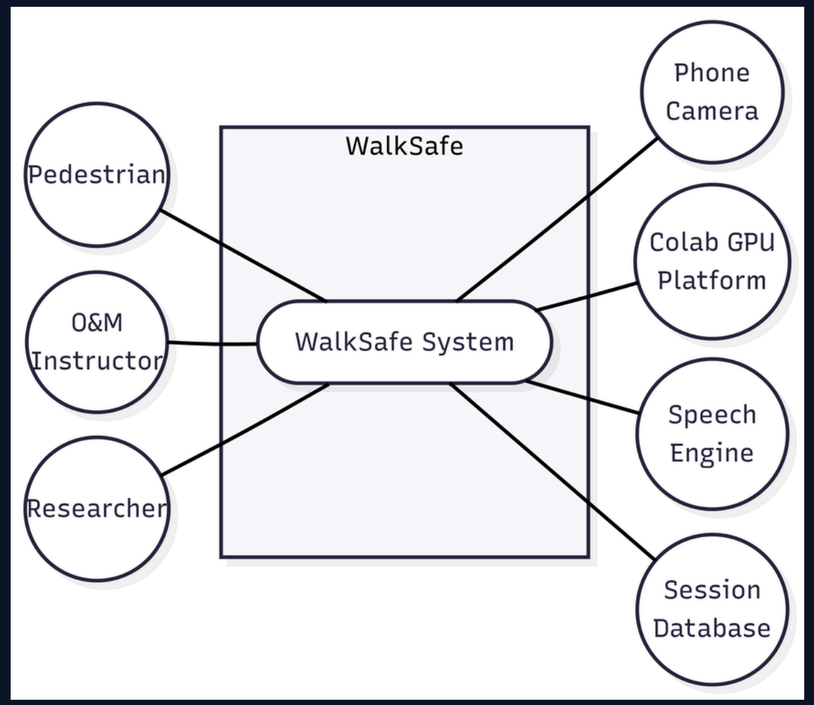
- Stop
- Yield
- Slow Down
- Walk Slow
- Walk Forward
- Sidestep Left
- Sidestep Right
- Turn Left
- Turn Right

Capability Levels

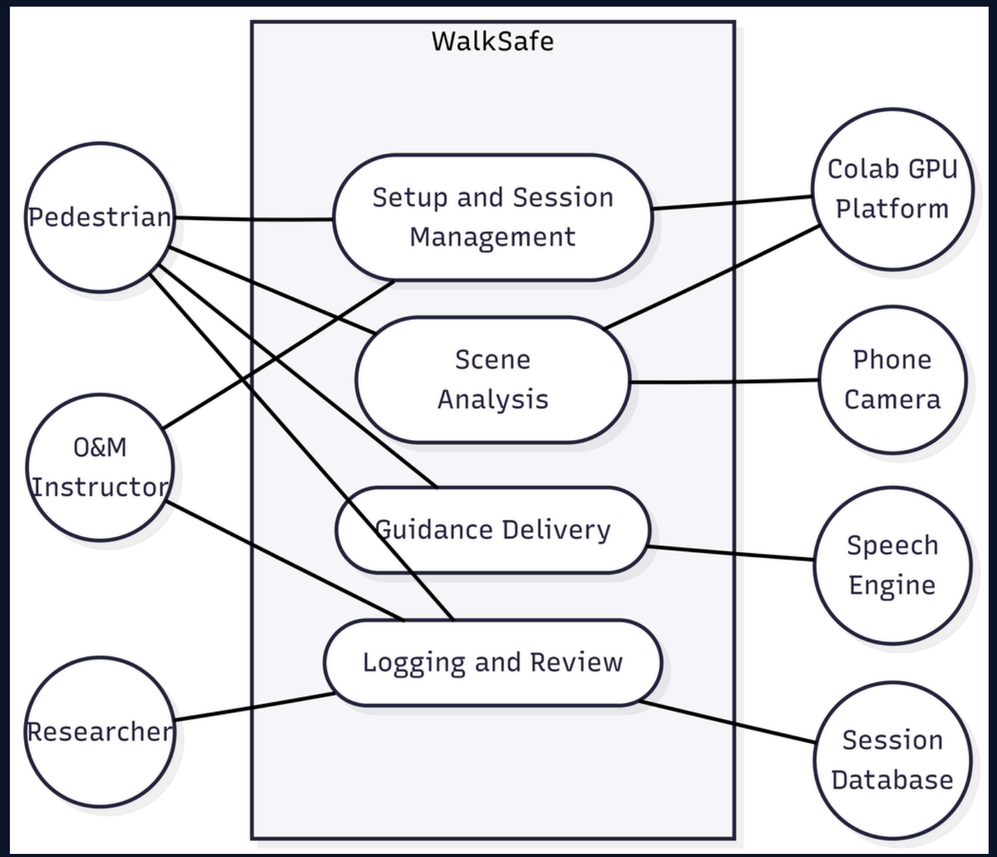
L0	**Everything working** | Full command vocabulary
L1	**Poor visibility** ig. glare, rain, blur | Cautious commands only; announce reduced confidence
L2	**Server unreachable** | Obstacle warnings only; announce loss of route guidance
L3	**On-phone check unavailable** | Announce that guidance has stopped; issue none
L4	**Battery exhausted or user ends session** | Announce shutdown with reason

Usecase Diagram : Level 0 & 1

Level 0

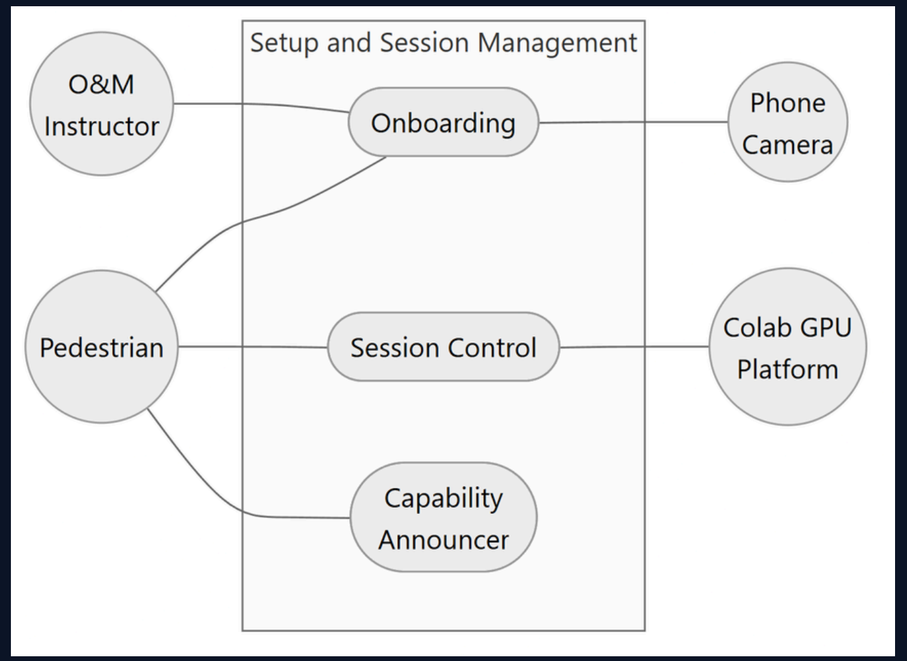


Level 1

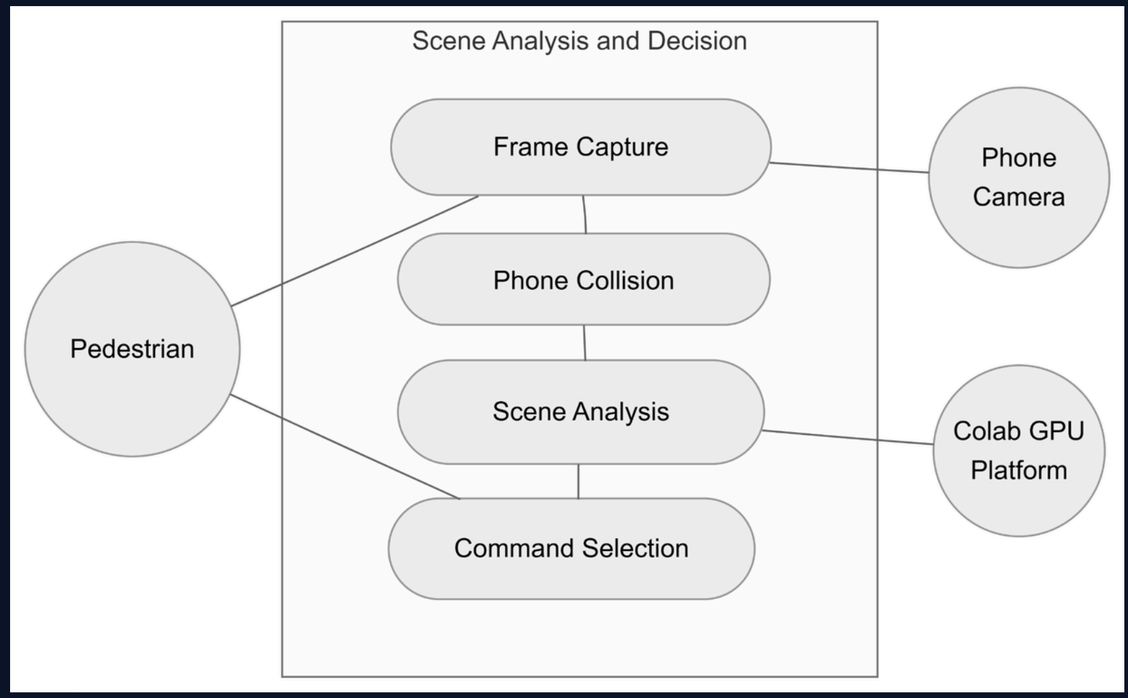


Usecase Diagram : Level 1.1 & 1.2

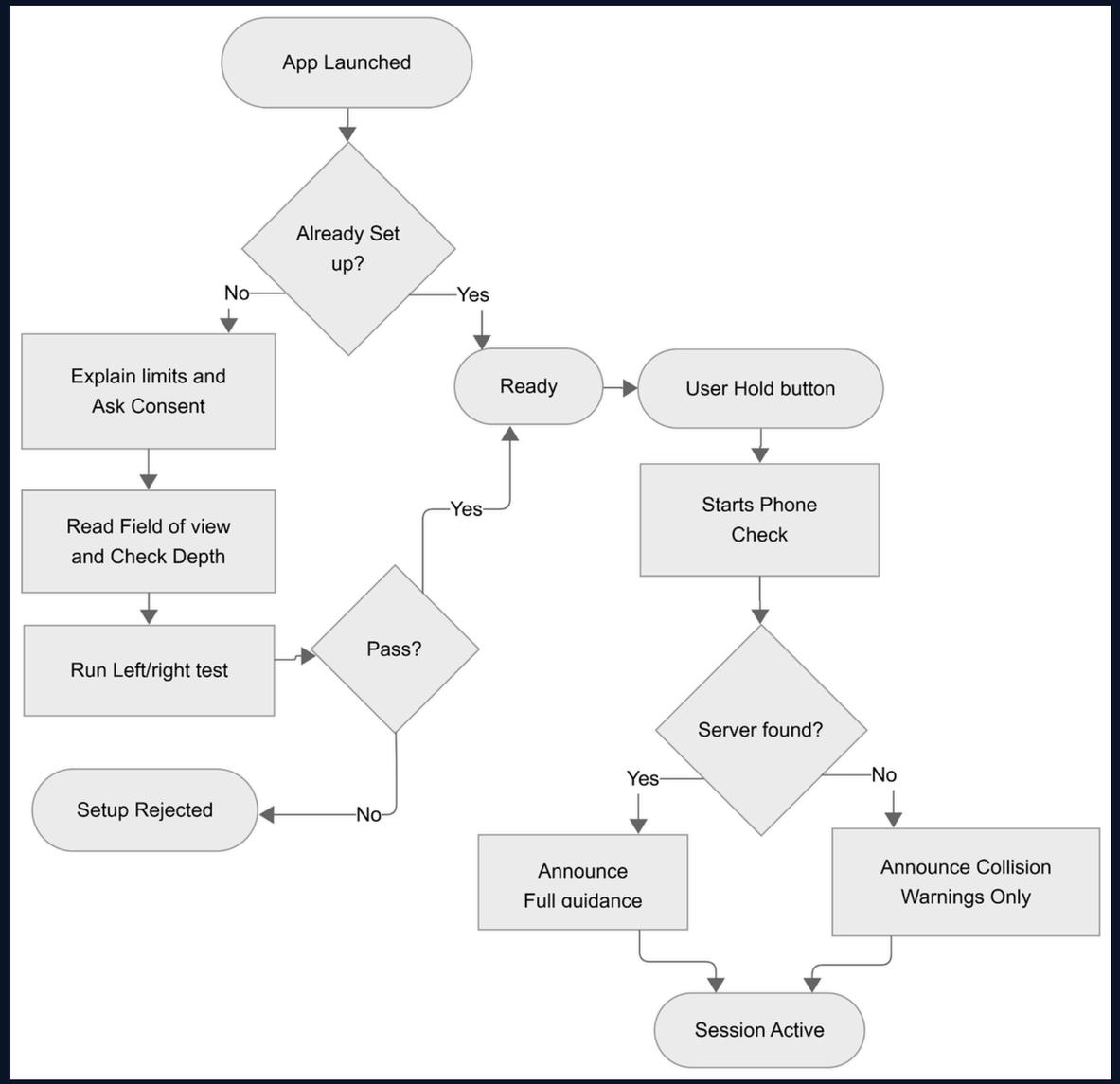
Level 1.1



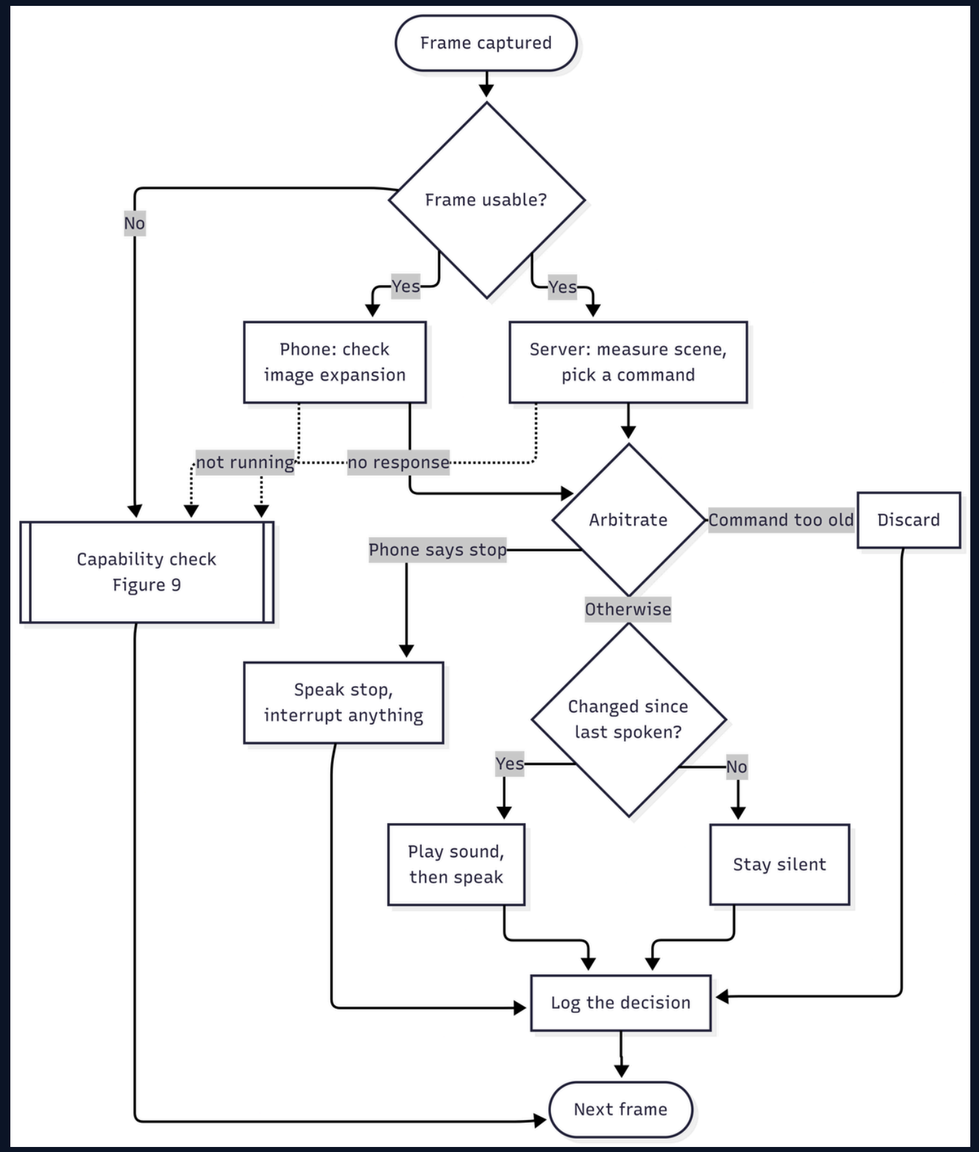
Level 1.2



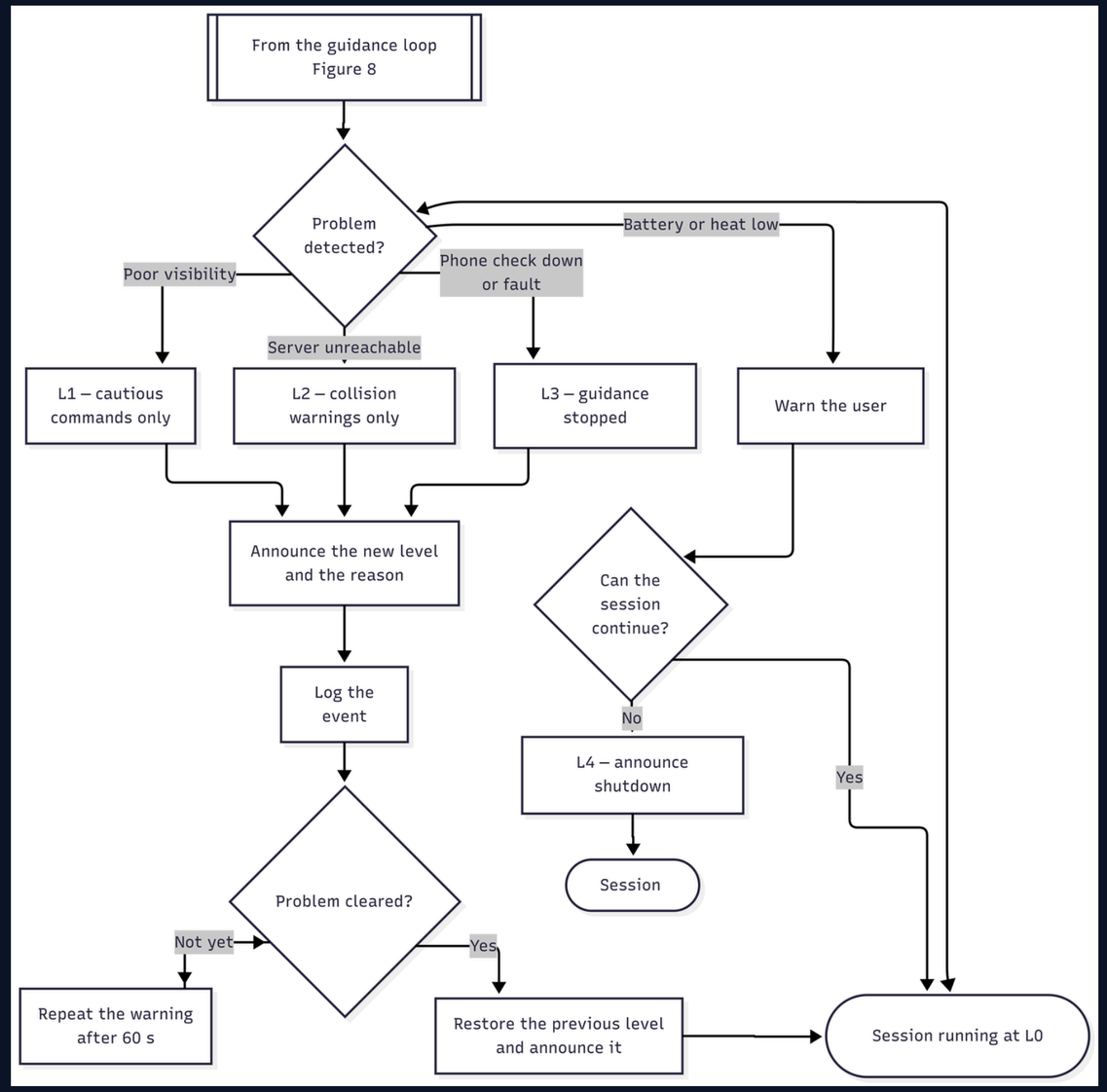
Activity Diagram : Setup & Session Start



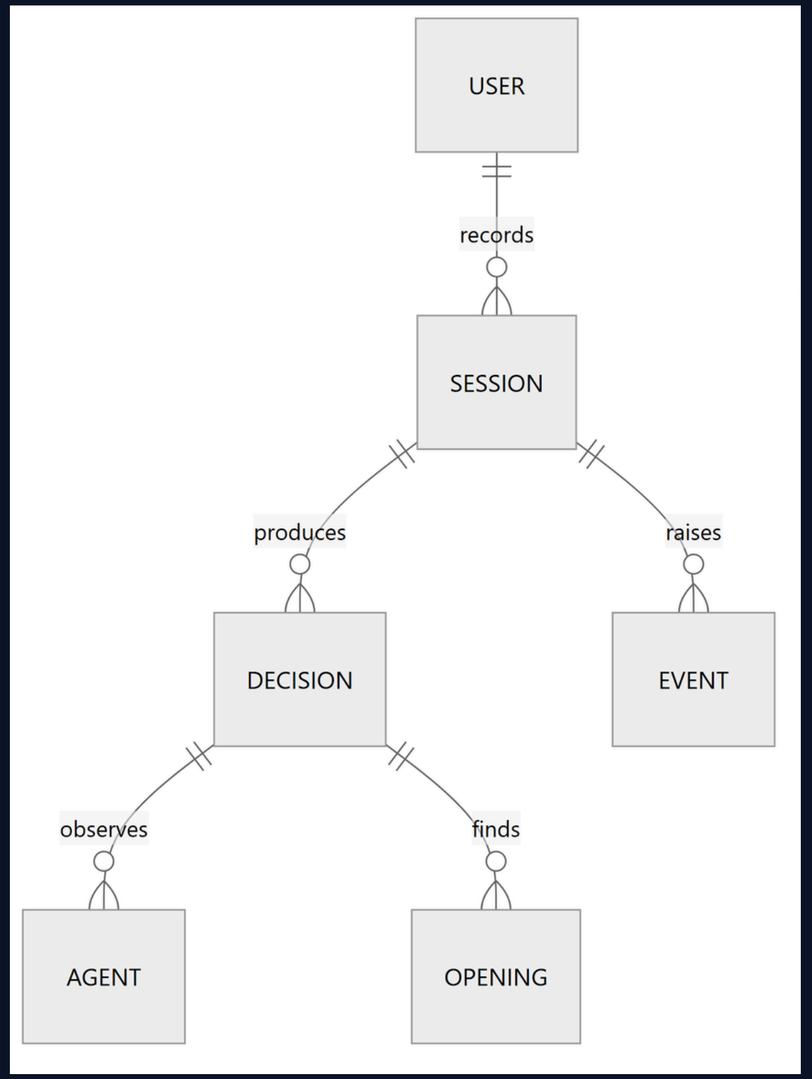
Activity Diagram : Guidance Loop



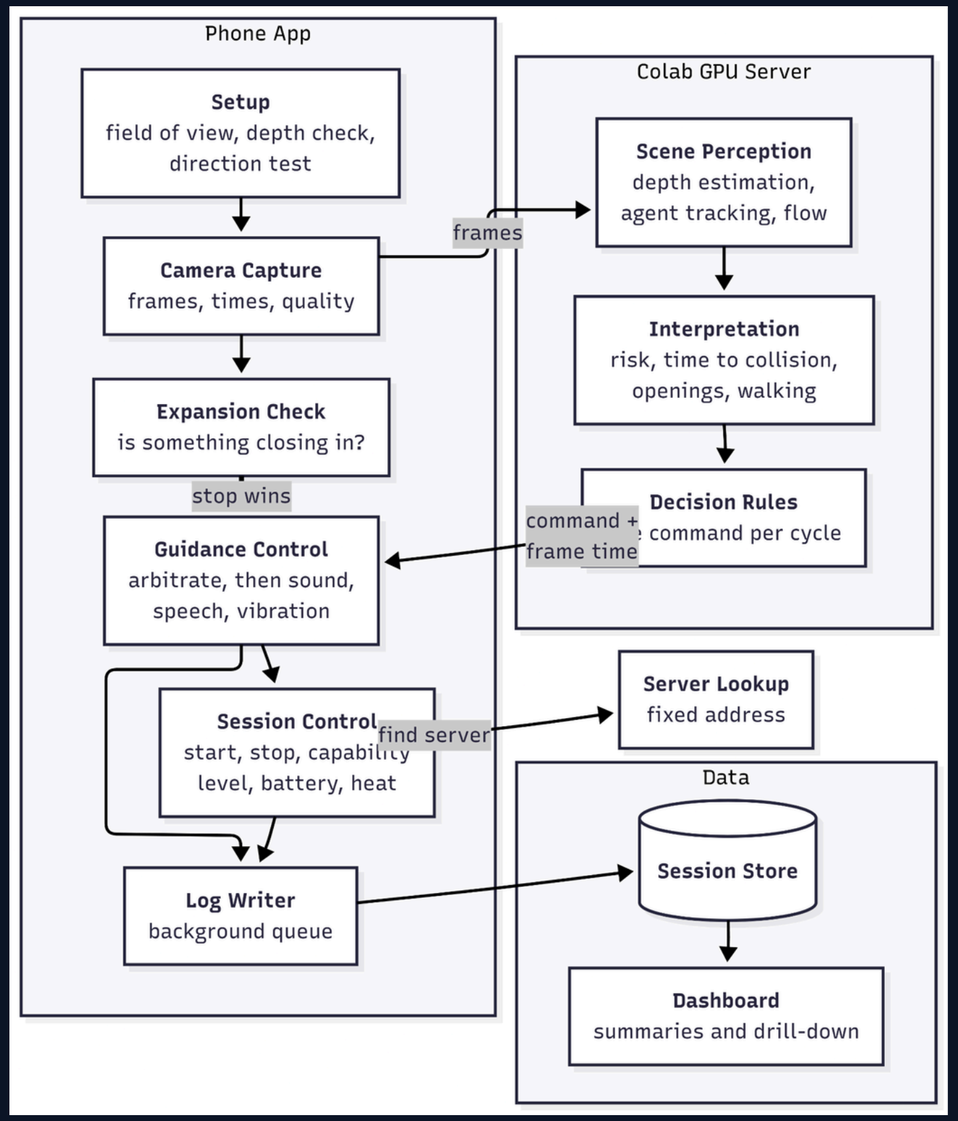
Activity Diagram : Capability Change and Recovery



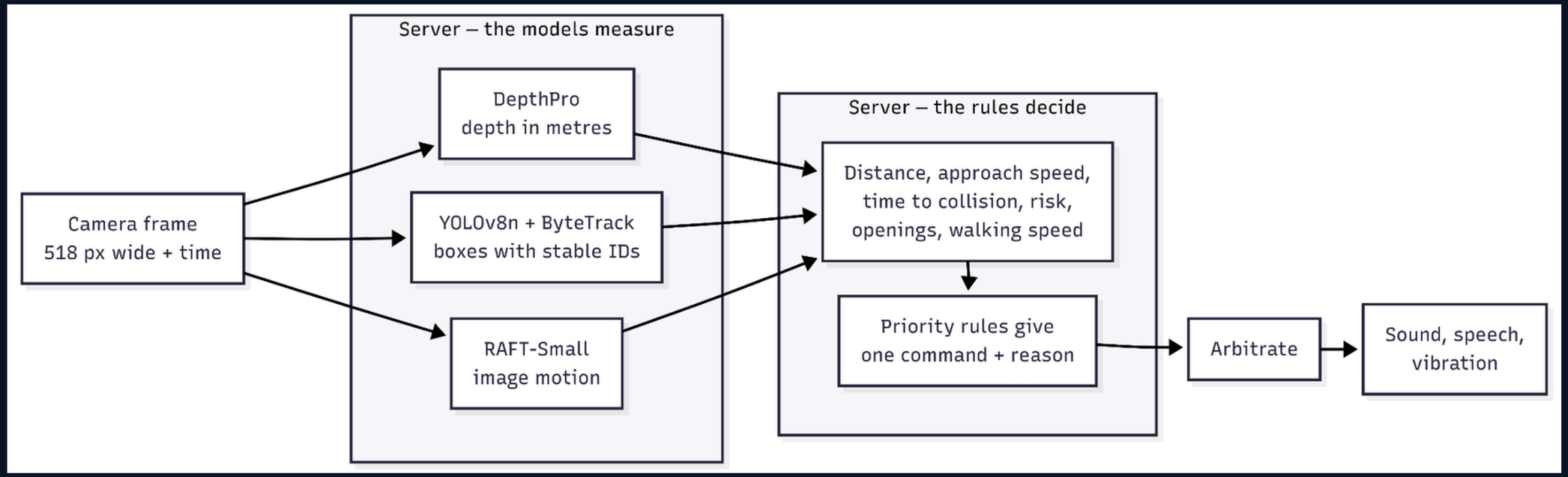
ER Diagram


USER
records
SESSION
produces
raises
DECISION
EVENT
observes
finds
AGENT
OPENING

Component Diagram



AI Pipeline


Server – the models measure
DepthPro
depth in metres
YOLOv8n + ByteTrack
boxes with stable IDs
RAFT-Small
image motion
Camera frame
518 px wide + time
Server – the rules decide
Distance, approach speed,
time to collision, risk,
openings, walking speed
Priority rules give
one command + reason
Arbitrate
Sound, speech,
vibration

Status & What’s Next

#	Deliverable	Status
D1	**Working prototype** — depth, detection and motion together on a Colab GPU; rule-based decision engine; browser interface	✅ **Complete**
D2	**Logging and dashboard** — SQLite record of every decision and the values behind it; command, risk and speed summaries	🔄 **In progress**
D3	**Smartphone application** — capture, on-phone collision check, server guidance, speech; spoken setup; screen-reader support	🔄 **In progress**
D4	**Test suite and test clips** — automated tests for the safety requirements, plus clips with known distances	⏳ **Planned**
D5	**Final report and demonstration**	⏳ **Planned**

Key Technologies & Tools

Language
Python

Text-to-Speech
edge-tts (fully offline)

Object Detection
YOLOv8n (Ultralytics)

Camera Interface
OpenCV

Multi-Object Tracking
ByteTrack

Deep Learning
PyTorch

Depth Estimation
Apple Depth Pro

Version Control
Git, GitHub

Motion Analysis
RAFT-Small (Optical Flow)

Demo Platform
Phone Camera with Colab Inference

References

[1] Z. Teed and J. Deng, “RAFT: Recurrent All-Pairs Field Transforms for Optical Flow,” ECCV 2020.

[2] L. Zhang et al., “ByteTrack: Multi-Object Tracking by Associating Every Detection Box,” ECCV 2022.

[3] Apple ML Research, “Depth Pro: Sharp Monocular Metric Depth in Less Than a Second,” arXiv:2410.02073, 2024.

[4] G. Jocher et al., “Ultralytics YOLOv8,” GitHub Repository, 2023.

[5] World Health Organization, “Blindness and Vision Impairment,” WHO Fact Sheet, 2023.

[6] H. Ye, Y. Chen and Z. Liu, “Sound of Vision: Natural Soundscape Synthesis for the Visually Impaired,” ICCV 2021.

Thank You

Any Questions?



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